



Comparative analysis of fecal microbiome, metabolome, and serum metabolome in cows with different milk urea nitrogen phenotypes

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ABSTRACT

Milk urea nitrogen, the primary form of nonprotein nitrogen (N) in milk, is an indirect indicator of N metabolism in dairy cows. The MUN concentrations are modulated by various factors, including dietary composition, physiological status, and environmental conditions. However, the potential roles of host gut microbiota and metabolome in the development of distinct MUN phenotypes remain insufficiently elucidated. Here, we compared fecal microbiota and fecal/serum metabolomes of high-MUN (HMUN) and low-MUN (LMUN) cows (n = 7 per group) under uniform feeding management using 16S rRNA gene sequencing and liquid chromatography-MS-based metabolomics. Compared with that of LMUN cows, the feces of HMUN cows exhibited relatively higher abundances of *UCG-009*, *UCG-002*, and *Christensenellaceae_R-7_group* and lower abundances of *Succinivibrio*, *Lachnospiraceae_NK3A20_group*, *Acetitomaculum*, *Prevotellaceae_UCG-001*, and *norank_f_Bifidobacteriaceae*. Metabolomic analysis revealed that HMUN cows had relatively lower levels of hydroxypropionic acid, N-myristoyl arginine, and N-eicosapentaenoyl tryptophan in feces, and reduced amounts of L-serine, linoleic acid, and butyrate in serum. Kyoto Encyclopedia of Genes and Genomes pathway mapping revealed that the metabolites relatively elevated in LMUN cows were primarily involved in the β -alanine metabolism pathway. The *UCG-009* and *Christensenellaceae_R-7_group* were positively correlated with MUN, whereas *Succinivibrio*, *Lachnospiraceae_NK3A20_group*, *Anaeroplasm*, and *Acetitomaculum* were negatively correlated. These microbial taxa were also significantly associated with

several fatty and AA metabolites, including adipic acid, undecenoic acid, dodecanedioic acid, DL-tryptophan, and L-leucine. Collectively, this study revealed certain differences in the gut microbiota and metabolome between cows with high and low MUN levels. These findings suggest that alterations in gut microbial composition and metabolic profiles may contribute to variations in MUN phenotypes and provide new insights into host factors influencing MUN metabolism in dairy cows.

Key words: milk urea nitrogen, fecal microbiota, fecal metabolome, serum metabolome, dairy cow

INTRODUCTION

Dairy animals, an important component of the agricultural economy, provide humans with diverse dairy products and contribute substantially to meat production. Cows, as the primary dairy species, contribute to over 80% of the global dairy product output (FAO, 2022). Over time, efforts to increase milk yield and improve milk quality have resulted in considerable demand for both the quantity and quality of protein in dairy production systems. In practice, particularly in countries or regions with less developed dairy industries, producers predominantly use relatively high-protein diets to meet the nutritional requirements of high-yielding cows. However, owing to the inefficiency of converting dietary N into productive outputs in dairy cows, excess protein intake increases N excretion through urine and feces (Castillo et al., 2000; Aguerre et al., 2010; Groenestein et al., 2019). This increases feeding costs and exacerbates the N footprint in the environment.

Milk urea nitrogen, the major form of nonprotein N in milk, is widely used as an indicator of N utilization efficiency in dairy cows. The MUN concentration generally increases with dietary protein supply (Nousiainen et al., 2004) and thus serves as a practical biomarker for evaluating the balance between N intake and utilization.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

In addition to dietary protein, energy availability (particularly the supply of NFC) also plays an important role in modulating MUN concentrations (Zhao et al., 2025c). Adequate energy supply promotes rumen microbial growth and microbial protein synthesis, thereby improving the capture of ruminal ammonia and enhancing overall N utilization efficiency. Consequently, monitoring MUN provides an integrated assessment of dietary protein adequacy and the nitrogen–energy balance in dairy cow feeding strategies.

In addition to nutritional factors, MUN concentrations are also influenced by nondietary variables such as season, ambient temperature, and physiological status of the animal (Carlsson and Pehrson, 1993; Trevaskis and Fulkerson, 1999; Aguilar et al., 2012; Bittante, 2022). Recently, growing evidence has highlighted the role of gastrointestinal microbiota in shaping host phenotypes. In cows, differences in rumen microbial composition, functional capacity, and metabolic activity have been consistently associated with milk protein yield, feed intake and efficiency, methane emissions, and other performance indicators (Li et al., 2019; Xue et al., 2020; Yang et al., 2024; Huang et al., 2025). Moreover, targeted nutritional modulation strategies have been shown to regulate rumen microbial structure and function, leading to improved ruminal N metabolism and enhanced N utilization efficiency (Matthews et al., 2019; Zhang et al., 2025a,b; Zhao et al., 2025a).

Rumen is the primary site of microbial fermentation and protein synthesis and plays a crucial role in N utilization in ruminants, related studies suggest that the lower gastrointestinal tract also contributes to host adaptation to diets differing in protein concentration (Rist et al., 2013; Tobón-Cornejo et al., 2025). Changes in dietary protein supply are frequently accompanied by concomitant shifts in carbohydrate availability, which can further modify substrate utilization patterns and shape gut microbial community structure (Wu et al., 2022). Through the degradation and assimilation of nitrogen-containing compounds, as well as interactions with host-derived urea N entering the intestinal lumen, gut microbiota may participate in postruminal N metabolism. These microbial activities have the potential to influence systemic N partitioning between productive pathways and excretory routes, thereby affecting circulating urea N concentrations and, ultimately, MUN levels. In addition, studies on microbial translocation have suggested that intestinal bacteria can be taken up by dendritic cells and transported via mesenteric lymph nodes to the mammary gland, with possible implications for mammary tissue function and metabolic regulation (Young et al., 2015). Emerging findings from research on the gut–brain and gut–mammary axes further suggest a potential regulatory role of the gut microbiota in host metabolic homeostasis (Rain-

ard, 2017; Almahal et al., 2025). Nevertheless, compared with the well-characterized role of the rumen microbiota, the contribution of gut microbial communities and their metabolic activities to individual variation in MUN phenotypes remains poorly understood.

To investigate whether gut microbial communities and associated metabolic profiles differ between dairy cows with contrasting MUN levels, milk, feed, fecal, and blood samples were collected from healthy multiparous lactating cows maintained under uniform feeding and management conditions. Cows with consistently high or low MUN levels over a 3-d period were selected for analysis. Fecal microbiota composition was characterized using 16S rRNA gene sequencing, and untargeted metabolomics was applied to both fecal and serum samples to profile host metabolic variations. These data will provide further insights into the gut microbial and metabolic characteristics of cows with divergent MUN phenotypes, offering a potential foundation for exploring the biological mechanisms underlying individual variation in MUN.

MATERIALS AND METHODS

Animals, Sample Collection, and Analysis

The experimental protocol was approved by the Animal Use Committee of the Institute of Animal Science, Chinese Academy of Agricultural Sciences (Beijing, China; protocol no. IAS2025–124). This study was conducted on a commercial dairy farm, where milk, feed, blood, and fecal samples were collected from multiparous lactating Chinese Holstein cows. Cows with similar parity (3.2 ± 1.2) and DIM (119 ± 18 d) were selected for the identification of contrasting MUN phenotypes. All cows were housed in the same freestall barn under identical feeding and management conditions, with ad libitum access to feed and water.

The TMR consisted of concentrates (rolled corn, soybean meal, and beet pulp) and forages (oat hay, alfalfa hay, and corn silage), as detailed in Supplemental Table S1, see Notes. The TMR was offered thrice daily at 0930, 1630, and 2230 h, and cows were milked at 1000, 1700, and 2300 h. Dry matter intake and milk yield were recorded over 3 consecutive days. Milk samples were collected at morning, afternoon, and evening milkings each day, mixed in a 4:3:3 ratio, and immediately transported to the laboratory for milk composition analysis. Seven cows with relatively high MUN levels (HMUN, 16.7 ± 0.72 mg/dL) and 7 cows with low MUN levels (LMUN, 11.3 ± 0.57 mg/dL) were selected for subsequent feed, blood, and fecal sampling based on the mean MUN concentrations measured over a 3-d period, with inclusion restricted to cows with consistently stable MUN levels throughout the sampling period.

The DMI was calculated based on the daily amounts of TMR offered to and refused by each cow, along with the corresponding DM content. Feed samples were collected at the beginning, middle, and end of TMR discharge from the mixer, combined thoroughly, and ~500 g was obtained as a composite sample using the quartering method. These samples were dried at 65°C for 24 h and ground using a 40-mesh grinder. Finally, DM content was determined at 105°C following AOAC methods (AOAC International, 2016).

Blood samples were collected from the coccygeal vein using sterile blood collection systems and 9-mL Vacutainer tubes after the completion of morning milking and ~1.5 h after morning feeding. The anticoagulated samples were transported on ice and analyzed immediately for hematological parameters. Nonanticoagulated samples were allowed to clot at room temperature for ~30 min and then centrifuged at $1,500 \times g$ for 15 min at 4°C to obtain the serum, which was temporarily stored at -20°C. Fecal samples were collected at the same time point as blood sampling, directly from the rectum using sterile gloves, placed into 5-mL cryovials, and immediately immersed in liquid nitrogen. Gloves were changed between animals to prevent cross-contamination. All frozen samples were transported to the laboratory on dry ice and stored at -75°C until further analysis.

Milk fat, protein, and lactose concentrations were determined using a mid-infrared milk analyzer (MilkoScan 133B, Foss Electric, Hillerød, Denmark). Milk N excretion was calculated as follows:

$$\text{milk N (g/d)} = \text{milk protein yield (g/d)} / 6.38.$$

The MUN yield was calculated as

$$\text{MUN yield (g/d)} = \text{milk yield (kg/d)} \times \text{MUN (mg/dL)} / 100.$$

Hematological indices (e.g., white and red blood cell counts) were measured using an automated hematology analyzer (HemaVet, Drew Scientific). The biochemical parameters of the blood (e.g., total protein and blood urea) were analyzed using an automated biochemical analyzer (TBA-FX8, Canon Medical Systems, Japan).

Fecal and Serum LC-MS Metabolomics Analysis and Data Processing

Approximately 100 mg of fecal material or 100 μL of serum was transferred into centrifuge tubes. Ground beads (when required) and extraction solvents, such as methanol–water (4:1, vol/vol) or acetonitrile–methanol (1:1, vol/vol), were added along with an internal standard (L-2-chlorophenylalanine, 0.02 mg/mL). The mixture

was briefly vortexed and homogenized using a cryogenic tissue disruptor for 6 min (-10°C, 50 Hz for solid matrices). Samples were then subjected to sonication under chilled conditions (5°C, 40 kHz) for 30 min to enhance metabolite release. After standing at -20°C for 30 min, the extracts were centrifuged ($13,000 \times g$, 15 min, 4°C). The resulting supernatant was either analyzed directly or evaporated under nitrogen and reconstituted in acetonitrile: water (1:1, vol/vol). A short ultrasonic step (5 min, 5°C, 40 kHz) and an additional centrifugation ($13,000 \times g$, 10 min, 4°C) ensured clarification before being transferred into liquid chromatography (LC) vials with inserts. To ensure analytical reproducibility, the pooled quality control samples were prepared by combining equal aliquots from every 5 to 10 experimental samples.

Metabolomic profiling was performed using a UHPLC system (Thermo Fisher Q Exactive HF-X) equipped with an HSS T3 column operating in both positive and negative ionization modes. Raw spectral data were processed using Progenesis QI (Waters, Milford, MA), which included baseline correction, peak detection, integration, retention-time alignment, and peak normalization. The resulting data matrix contained retention time, m/z , and normalized intensities. Compound annotation was achieved by matching the MS and MS/MS spectra against multiple databases, including Human Metabolome Database (HMDB; <http://www.hmdb.ca/>), METLIN (<https://metlin.scripps.edu/>), and an in-house spectral library, to ensure high-confidence metabolite identification.

Fecal DNA Extraction and Library Construction

Total DNA was isolated from fecal material using the E.Z.N.A. DNA Kit (Omega Bio-tek, Norcross, GA) following the manufacturer's protocol. The DNA integrity was assessed using 1% agarose gel electrophoresis, and DNA concentration and purity were measured using a NanoDrop 2000 spectrophotometer (Thermo Scientific). For microbial community profiling, the V3–V4 region of the 16S rRNA gene was amplified using barcode primers. The PCR was performed under the following conditions: initial denaturation at 95°C for 3 min; 27 cycles of 95°C for 30 s, 55°C for 30 s, and 72°C for 30 s; followed by a final extension at 72°C for 10 min. Each 20 μL reaction contained $5 \times$ FastPfu buffer, dNTP, primers, FastPfu DNA polymerase, template DNA, and nuclease-free water. The PCR amplicons were separated on 2% agarose gels, excised, and purified using a commercial gel extraction kit (PCR Clean-Up Kit, Yuhua, China). The DNA concentration was quantified using a Qubit 4.0 fluorometer (Thermo Fisher Scientific). Sequencing libraries were prepared using the NEXTFLEX Rapid DNA-Seq Kit, which included adapter ligation, size selection using

magnetic beads, PCR enrichment, and purification. The resulting libraries were then sequenced.

16S rRNA Gene Sequencing and Data Processing

Sequencing of the 16S rRNA gene was conducted using the Illumina MiSeq PE300 platform (Shanghai Meiji Bio-Pharmaceutical Technology Co. Ltd., Shanghai, China), and the resulting raw reads were submitted to the National Center for Biotechnology Information Sequence Read Archive (NCBI SRA) database. Raw data were subjected to quality filtering using FASTP to remove low-quality bases and ambiguous nucleotides. Paired-end reads were merged using FLASH and sequences were assigned to individual samples according to their unique barcodes. To minimize bias from sequencing depth, high-quality reads were denoised using the DADA2 algorithm in QIIME2 to yield amplicon sequence variants (ASV). To reduce the effect of uneven sequencing depth on downstream α - and β -diversity analyses, all samples were rarefied to an equal depth of 20,000 sequences per sample. After rarefaction, the average Good's coverage across samples remained 99.09%, indicating sufficient sequencing depth. The taxonomic assignment was performed using the SILVA 16S rRNA reference database (<http://www.arb-silva.de/>). In addition to taxonomic profiling, microbial diversity metrics (α - and β -diversity) were calculated in QIIME2 to assess within- and between-sample variability. Functional predictions of microbial communities were obtained using PICRUST2 (<https://github.com/picrust/picrust2>), and Bray–Curtis dissimilarity matrices were generated for multivariate statistical analyses such as principal coordinate analysis (PCoA).

Spearman Correlation Analysis

To evaluate the relationships between microbial composition, metabolites, and MUN levels, Spearman's rank correlation was applied, and outcomes with P -values < 0.05 were considered significant. All correlation analyses and visualizations were conducted using the R packages ggcorrplot (Kassambara, 2019) and MetaNet (Peng et al., 2025). The ggcorrplot package was used to generate correlation heat maps, providing a clear visualization of the relationships between variables. Additionally, the MetaNet package was used to construct and analyze metabolism-related networks, facilitating the exploration of interactions between the fecal microbial community and metabolites.

Statistical Analysis

The DMI, milk yield, milk composition, and blood parameters were compared using SAS 9.4 (SAS Insti-

tute Inc., Cary, NC). Differences in group means were assessed using independent t -tests, with statistical significance defined as $P < 0.05$, and P -values between 0.05 and 0.10 were considered indicative of a tendency or trend toward a difference.

Microbial sequencing data were processed using the Meiji Cloud platform (<https://cloud.majorbio.com>). Alpha diversity (Chao1 and Shannon) was estimated using mothur (<http://www.mothur.org/wiki/Calculators>), and group differences were assessed using the Wilcoxon rank-sum test. Community structure was visualized with PCoA based on the Bray–Curtis distance, and the relative abundances of bacterial taxa were compared between groups using the Wilcoxon test. Differential taxa were identified using linear discriminant analysis effect size (LEfSe; linear discriminant analysis [LDA] > 2.0 , $P < 0.05$).

Metabolomic data were processed and analyzed using the ropls R package (Thévenot et al., 2015). Principal component analysis (PCA) and orthogonal projection to latent structures discriminant analysis were performed to visualize group separation. Differential metabolites were identified using Wilcoxon rank-sum tests, with compounds meeting variable importance in projection (VIP) > 1 , $P < 0.05$, and $|\text{fold change (FC)}| > 1$ considered indicative of potential group differences. Metabolite annotation was based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, and pathway enrichment analysis was conducted using Python's scipy stats module.

RESULTS

Feed Intake, Milk Yield, Milk Composition, and Blood Biochemical Parameters

Tables 1 and 2 present a comparison of the DMI, milk yield, milk composition, and blood parameters between HMUN and LMUN cows. Blood urea concentrations were significantly higher in the HMUN group ($P < 0.05$), consistent with the MUN results. Additionally, the red blood cell content in the HMUN group was higher than that in the LMUN group ($P < 0.05$). No significant differences were observed between the 2 groups in any of the other measured blood indicators ($P > 0.05$).

Fecal Microbial Community Composition and Diversity

Approximately 1,549,835 raw sequences were generated from 14 fecal samples. After denoising and quality filtering, 731,614 high-quality sequences were retained for downstream analysis (Supplemental Table S2, see Notes). Microbial richness was evaluated using the Chao

Table 1. Analyzed DMI, milk production, and milk composition between HMUN and LMUN cows

Item ¹	HMUN	LMUN	SEM	P-value
DMI (kg/d)	21.7	22.6	1.19	0.60
Production				
Milk yield (kg/d)	34.1	35.4	1.59	0.58
Protein yield (kg/d)	1.04	1.16	0.05	0.10
Fat yield (kg/d)	1.48	1.51	0.07	0.74
MUN yield (g/d)	5.67	4.18	0.26	<0.01
Milk N (g/d)	163	181	7.27	0.10
Composition				
Protein (%)	3.07	3.28	0.10	0.15
Fat (%)	4.40	4.36	0.23	0.92
Lactose (%)	4.89	4.83	0.06	0.47
MUN (mg/dL)	16.7	11.3	0.25	<0.01

¹N = nitrogen.

and angiotensin-converting enzyme (ACE) estimators, while microbial diversity was assessed using the Shannon and Simpson indices. The Shannon and Simpson indices did not differ significantly between the HMUN and LMUN groups ($P > 0.05$). Although no significant differences were observed in the Chao and ACE indices, both indices were numerically lower in the LMUN group ($P = 0.074$; Supplemental Figure S1, see Notes). The annotation of ASV facilitated comprehensive characterization of the fecal microbiota, including its community composition and relative abundance. Among the 16 bacterial phyla identified, the most abundant were *Bacillota* (65.87%), *Bacteroidota* (29.46%), *Spirochaetota* (2.06%), and *Actinomycetota* (1.80%; Supplemental

Table S3, see Notes). Genus-level profiling identified the dominant taxa, which included UCG-005 (9.78%), *Romboutsia* (8.61%), *Rikenellaceae_RC9_gut_group* (8.36%), *norank_f_UCG-010* (6.19%), *unclassified_f_Lachnospiraceae* (5.61%), *Paeniclostridium* (5.60%), *Bacteroides* (5.50%), *Alistipes* (3.93%), *Prevotellaceae_UCG-003* (3.13%), and *Christensenellaceae_R-7_group* (2.64%; Supplemental Table S4, see Notes).

Fecal Microbial Community Differences in Cows with Different MUN Levels

To obtain a comprehensive overview of the gut microbial communities in cows with different MUN levels, β -diversity analysis was performed to compare the overall structure of the fecal microbiota between the HMUN and LMUN groups. The PCoA showed that no significant differences were detected at either the phylum or genus level ($P > 0.05$). Although the ordination plot displayed a visual tendency toward separation between the 2 groups, this observation was descriptive only and did not indicate a supported difference in community structure (Supplemental Figure S2, see Notes).

The relative abundances of fecal microbiota between the HMUN and LMUN groups were compared using the Wilcoxon rank-sum test combined with LefSe analysis. At the phylum level, *Fibrobacterota* ($P = 0.024$, LDA = 2.62) and *Elusimicrobiota* ($P = 0.011$, LDA = 2.61) were significantly enriched in the HMUN group, whereas *Pseudomonadota* showed numerically higher relative abundance in the LMUN group, although this difference did not reach significance ($P = 0.074$, LDA = not applicable; Figure 1A). Genus-level analysis revealed that *UCG-009* ($P = 0.002$, LDA = 3.56), *UCG-002* ($P = 0.007$, LDA = 3.23), and *Christensenellaceae_R-7_group* ($P = 0.021$) were significantly enriched in the HMUN group, whereas the LMUN group exhibited increased relative abundances of *Lachnospiraceae_NK3A20_group* ($P = 0.011$, LDA = 3.51), *Succinivibrio* ($P = 0.015$), *Acetivomaculum* ($P = 0.015$, LDA = 3.84), and *Prevotellaceae_UCG-001* ($P = 0.030$, LDA = 3.79; Figure 1B).

Feces and Serum Metabolic Profile of Cows in Different Groups

The PCA was used to summarize the metabolic landscapes of the HMUN and LMUN groups. The score plots of fecal and serum metabolites showed a partial separation between the 2 groups. The partial least squares discriminant analysis (PLS-DA) was performed to visualize potential differences. Although the cumulative R^2X and R^2Y values exceeded 0.5, the response permutation test yielded a Q^2 value close to zero, indicating limited predictive ability and confirming the absence of overfit-

Table 2. Analyzed blood hematological parameters and plasma metabolite markers between HMUN and LMUN cows

Item ¹	HMUN	LMUN	SEM	P-value
Blood hematological parameter				
White blood cell ($\times 10^9/L$)	6.81	6.63	1.00	0.86
Lymphocyte count ($\times 10^9/L$)	2.73	2.57	0.66	0.81
Neutrophil count ($\times 10^9/L$)	3.49	3.46	0.41	0.94
Red blood cell ($\times 10^{12}/L$)	6.21	5.81	0.17	0.03
Hemoglobin (g/L)	87.77	85.36	2.06	0.25
Hematocrit (%)	28.16	26.81	0.80	0.10
Platelet count ($\times 10^9/L$)	514.00	490.00	76.58	0.76
Platelet crit (%)	0.29	0.27	0.04	0.53
Plasma metabolite				
Total protein (g/L)	80.07	75.49	2.08	0.14
Albumin (g/L)	33.57	35.97	1.36	0.23
Globulin (g/L)	46.5	39.51	3.18	0.15
ALT (U/L)	28.94	26.52	3.53	0.64
AST (U/L)	119.66	95.49	13.35	0.22
AST/ALT	4.48	3.56	0.53	0.24
Glucose (mmol/L)	3.16	3.20	0.13	0.80
Cholesterol (mmol/L)	3.76	3.29	0.35	0.35
Triglycerides (mmol/L)	0.10	0.10	0.01	0.48
Urea (mmol/L)	4.50	3.74	0.25	0.05
Total bilirubin ($\mu\text{mol/L}$)	0.78	0.63	0.08	0.20
Direct bilirubin ($\mu\text{mol/L}$)	0.29	0.36	0.07	0.50
NEFA (mEq/L)	0.35	0.34	0.02	0.80

¹ALT = alanine aminotransferase; AST = aspartate aminotransferase; NEFA = nonesterified fatty acids.

ting (Figure 2). Therefore, the PLS-DA score plots are presented primarily for visualization, and interpretation of group separation is further supported by PCA and unsupervised analyses.

In total, 1,016 metabolites were identified in the fecal samples (Supplemental Table S5, see Notes) and 897 in the serum (Supplemental Table S6, see Notes). The most represented superclasses in the fecal metabolome were lipids and lipid-like molecules (35.26%), organic acids and their derivatives (16.54%), and benzenoids (9.74%). Within these categories, the predominant subclasses were AA, peptides, and analogs (12.31%), followed by fatty acids and conjugates (7.44%), bile acids, alcohols, and derivatives (3.33%), and carbonyl compounds (3.08%).

The analysis of cow serum samples revealed that organic acids and their derivatives (26.81%), lipids and lipid-like molecules (26.17%), organoheterocyclic compounds (16.81%), and aromatic compounds (12.34%) were the predominant metabolite superclasses. The main subclasses were AA, peptides, and analogs (14.89%), glycerophosphocholines (5.11%), and fatty acids and conjugates (4.89%; Supplemental Figure S3, see Notes).

Feces and Serum Differential Metabolites of Cows in Different Groups

Fecal and serum metabolites were compared between the HMUN and LMUN groups using a nonparametric

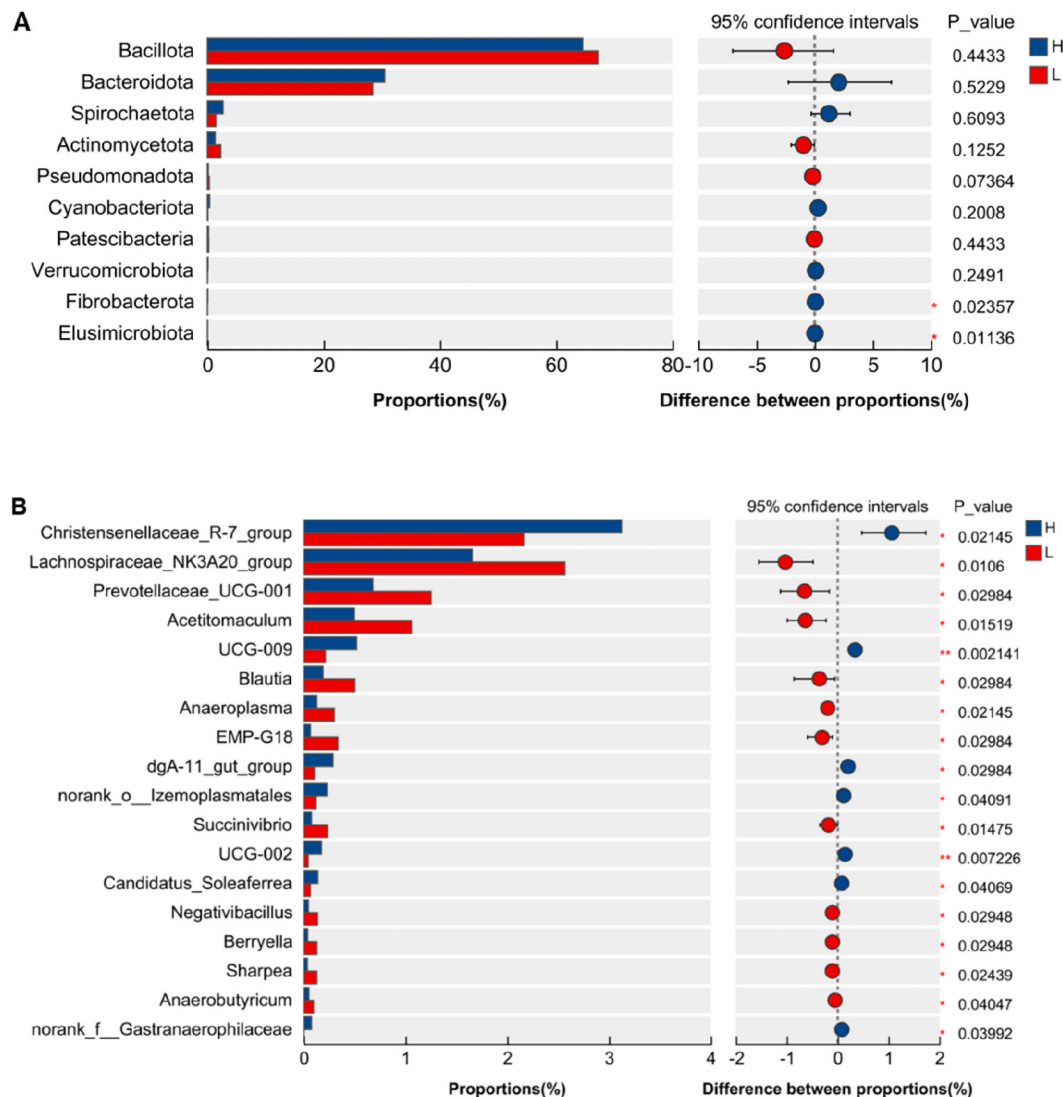


Figure 1. The composition of different microorganisms in feces samples between HMUN and LMUN cows at (A) the phylum level and (B) the genus level using the Wilcoxon rank-sum test. (C) Cladogram shows linear discriminant analysis effect size (LEfSe) analysis of different fecal microbiota between HMUN and LMUN cows from phylum to genus level.

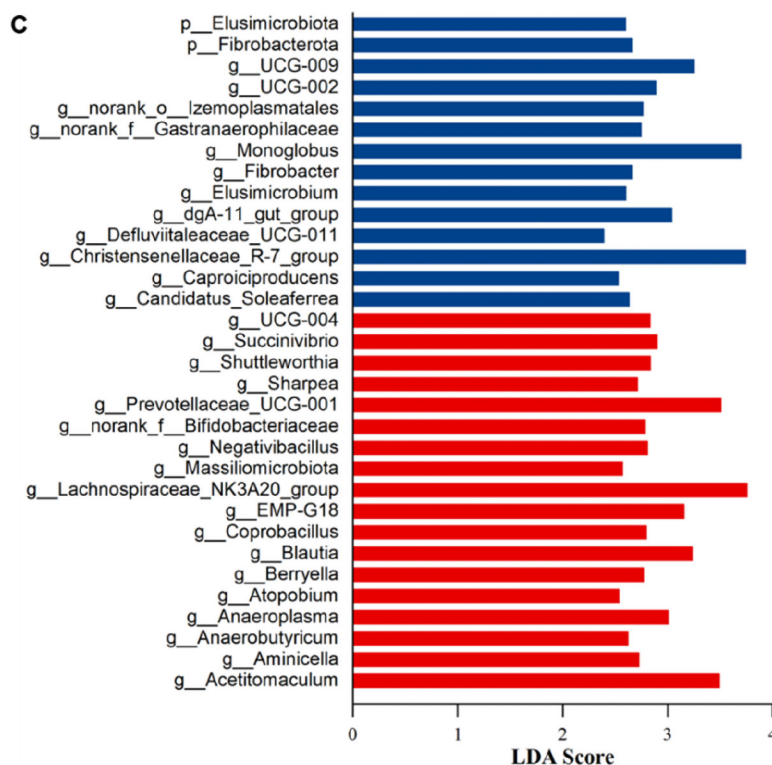


Figure 1 (Continued). The composition of different microorganisms in feces samples between HMUN and LMUN cows at (A) the phylum level and (B) the genus level using the Wilcoxon rank-sum test. (C) Cladogram shows linear discriminant analysis effect size (LEfSe) analysis of different fecal microbiota between HMUN and LMUN cows from phylum to genus level.

2-tailed Wilcoxon rank-sum test. Metabolites were classified as differentially abundant when they satisfied the thresholds of $P < 0.05$, $VIP > 1.0$, and $|FC| > 1$. In the fecal samples, 94 differential metabolites were identified, of which 64 were upregulated in the HMUN group, including hydroxytyrosol acetate, glycylproline, and kainic acid, and 30 were downregulated, including hydroxypropionic acid, N-myristoyl arginine, and N-icosapentaenoyl tryptophan (Supplemental Table S7, see Notes). The KEGG pathway analysis revealed that the metabolites elevated in the HMUN group were predominantly linked to α -linolenic acid, nucleotide, and purine metabolism. In contrast, those enriched in the LMUN group were associated with phenylalanine metabolism, purine metabolism, riboflavin metabolism, and β -alanine metabolism (Figure 3A).

In serum samples, 96 differential metabolites were detected, with 62 upregulated in the HMUN group, including 4-keto-azetidine, γ -glutamyl-glutamic acid, and 1-(4Z,7Z,10Z,13Z,16Z,19Z-docosaheptaenoyl)-sn-glycerol-3-phosphocholine and 34 downregulated, such as L-serine, linoleic acid, and butyrate (Supplemental Table S8, see Notes). The KEGG pathway mapping revealed that the elevated metabolites in HMUN cows were mainly associated with glycerophospholipid metabolism,

fat digestion, and absorption, as well as pathways linked to cutin, suberin, and wax biosynthesis. Conversely, metabolites enriched in LMUN cows were related to ABC transporters, β -alanine metabolism, protein digestion and absorption, and aminoacyl-tRNA biosynthesis (Figure 3B).

Relationship Among Microbiome, Metabolome, and MUN

To explore the associations among the gut microbiota, fecal and serum metabolites, and MUN concentrations, we used Spearman's rank correlation analysis. At the genus level, several taxa were significantly associated with MUN. The *UCG-009* and *Christensenellaceae_R-7_group* were aligned with higher MUN concentrations, contrary to *Succinivibrio*, *Lachnospiraceae_NK3A20_group*, *Prevotellaceae_UCG-001*, *Anaeroplasma*, and *Acetitomaculum*, which demonstrated negative associations (Figure 4A). When assessing the relationships between gut microbes and fecal metabolites, *Christensenellaceae_R-7_group*, *UCG-009*, and *UCG-002* showed significant positive associations ($r > 0.5$, $P < 0.05$) with adipic acid, deoxyguanosine, and undecylenic acid and negative associations with AMP and uridine monophosphate ($r < -0.5$,

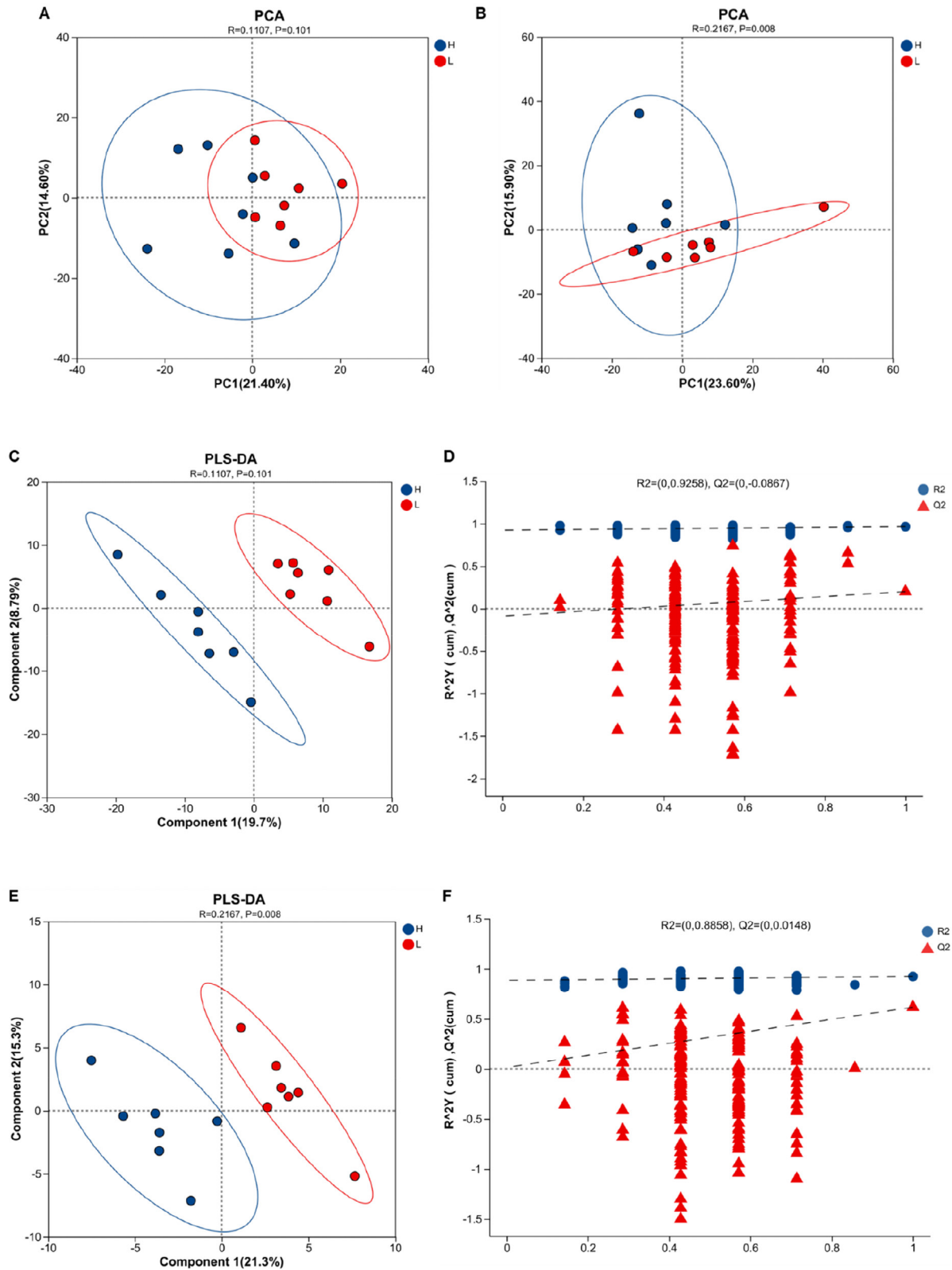


Figure 2. Principal component analysis (PCA) and partial least squares discriminant analysis (PLS-DA) score plots (A and C for fecal metabolites; B and E for serum metabolites) and PLS-DA permutation testing (D and F) between the HMUN and LMUN groups. The $R^2X(\text{cum})$ and $R^2Y(\text{cum})$ represent the cumulative interpretation rates for the X and Y matrices of the model, respectively. The Q^2 value indicates the predictive ability of the model, with Q^2 values near or below 0 confirming that the model is not overfitting. PC = principal component.

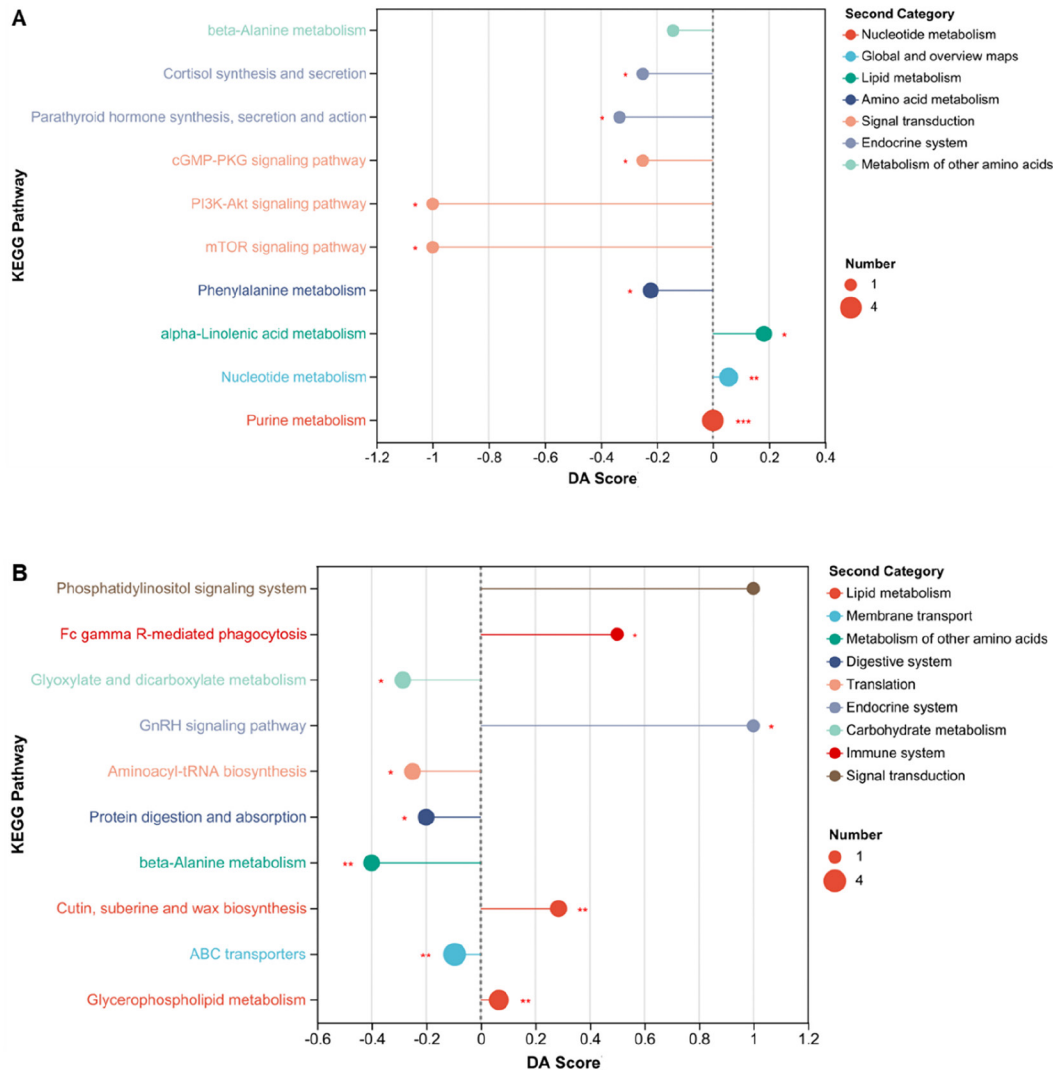


Figure 3. Metabolic pathways of fecal (A) and serum (B) differential metabolites between HMUN and LMUN cows. The horizontal axis shows the Differential Abundance Score (DA Score), and the vertical axis lists KEGG metabolic pathways. Circle size indicates the number of differentially expressed metabolites, with larger circles representing more metabolites. Circles on the right of the center axis, with longer lines, indicate upregulation of that pathway, while those on the left with longer lines suggest downregulation. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

$P < 0.05$). In contrast, *Succinivibrio*, *Anaeroplasma*, and *Acetitomaculum* showed the opposite correlation pattern, being inversely linked to deoxyguanosine, adipic acid, and undecylenic acid, but positively related to valtrate and midodrine ($r > 0.5$, $P < 0.05$). Additionally, *Lachnospiraceae_NK3A20_group* exhibited significant negative correlations with undecylenic, adipic, and dodecanedioic acids ($r < -0.5$, $P < 0.05$; Figure 4B).

Serum metabolite analysis revealed that *UCG-009* and *Christensenellaceae_R-7_group* positively correlated with 1-oleoyl-2-linoleoyl-sn-glycero-3-phosphocholine and 1-palmitoyl-2-linoleoyl-sn-glycero-3-PC ($r > 0.5$, $P < 0.05$), whereas their relationships with phosphatidylcholine and triethylamine were negative ($r < -0.5$, $P < 0.05$). Conversely, *Succinivibrio*, *Anaeroplasma*, and

Acetitomaculum positively correlated with phosphatidylcholine and triethylamine and negatively correlated with 1-oleoyl-2-linoleoyl-sn-glycero-3-phosphocholine and PC (18:3(9Z,12Z,15Z)/18:2(9Z,12Z)). Moreover, *Acetitomaculum* was also positively related to L-leucine and DL-tryptophan ($r > 0.5$, $P < 0.05$), and *Lachnospiraceae_NK3A20_group* positively correlated with phosphatidylcholine, triethylamine, and butyryl ($r > 0.5$, $P < 0.05$; Supplemental Figure S4, see Notes).

DISCUSSION

The MUN is an important indicator of dietary energy and protein balance and provides information on the metabolic status of dairy cows. Data from DHI reports show

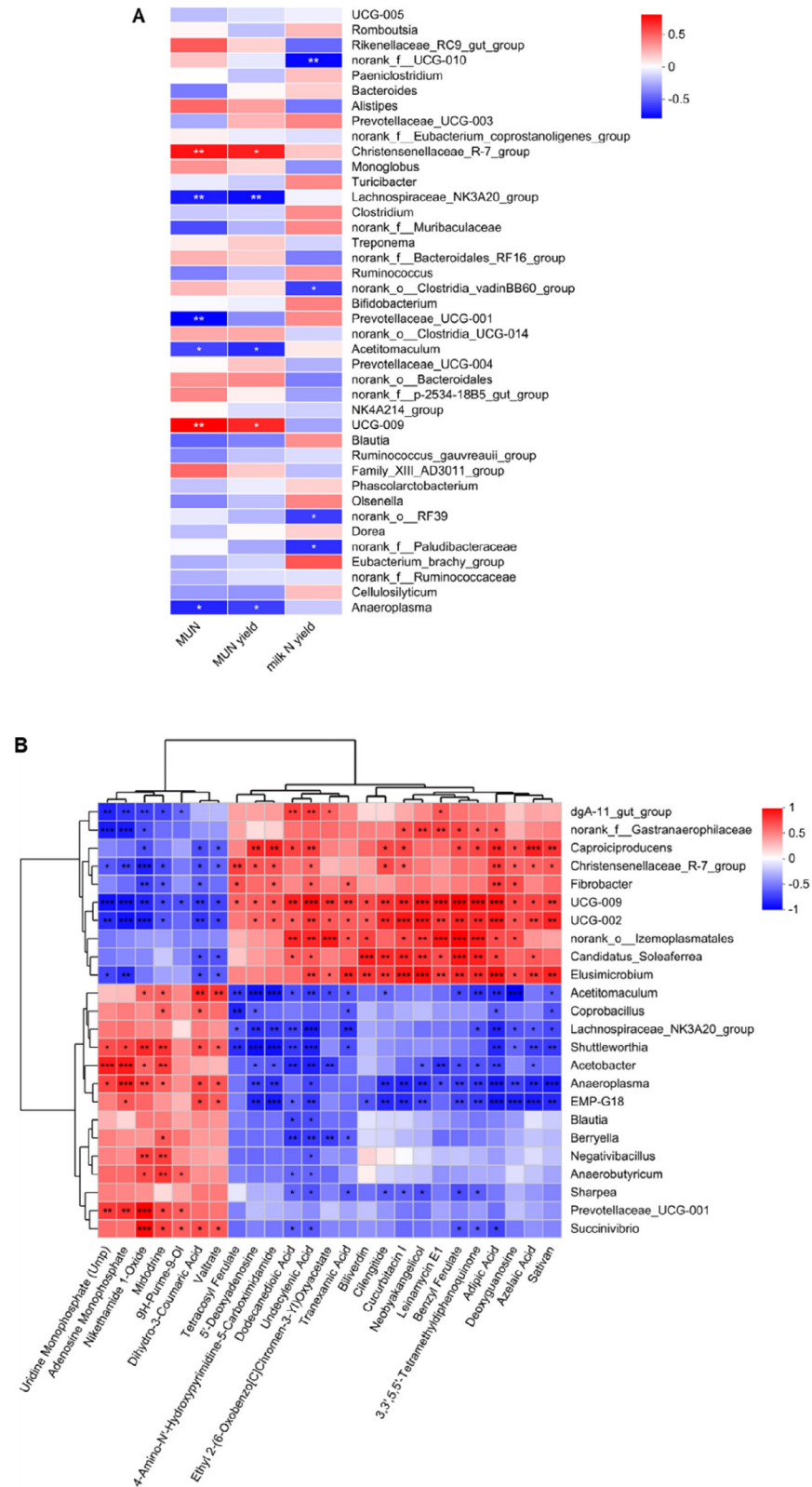


Figure 4. Interactions among fecal microbiota, fecal metabolome, and MUN. (A) Spearman's correlations between fecal microbiota and MUN. (B) Spearman's correlations between fecal microbiota and fecal metabolites. Red = positive, blue = negative correlations. Numbers indicate Spearman's correlation coefficients (r). Significant correlations ($|r| > 0.5$ and $P < 0.05$) are shown in bold. P -values were not adjusted for multiple testing. Correlation strength increases as $|r|$ increases. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

that, even under identical diets and management, MUN levels can vary substantially among cows within the same herd, suggesting inherent differences in N metabolism. Increasing evidence indicates that gastrointestinal microbiota influence animal productivity, and variations in the structure and function of the rumen microbiome are associated with differences in host phenotypes (Matthews et al., 2019; Xue et al., 2020; Zhang et al., 2024; Huang et al., 2025). A recent machine learning-based study identified 9 gut microbial taxa potentially linked to MUN levels (Yu et al., 2024). However, the relationships among gut microbiota, microbial metabolites, and host metabolism in relation to individual MUN variation remain largely unexplored. In this study, we conducted a preliminary investigation of fecal and blood microbial and metabolic profiles in cows with high- and low-MUN phenotypes. Among production performance, milk composition, and blood parameters, only MUN and plasma urea concentrations differed significantly between groups, whereas other measures showed no significant differences. These results suggest that, under uniform feeding and management conditions, variation in MUN may primarily reflect differences in N metabolism rather than broad physiological differences among cows.

The gut microbiota is an important regulatory factor closely linked to nutrient digestion and metabolism and can be influenced by diet, age, environment, and prebiotics. Taxonomic analysis revealed that *Bacillota*, *Bacteroidota*, *Spirochaetota*, and *Actinomycetota* were the predominant bacterial phyla in feces samples, consistent with previous reports in other ruminants (Wang et al., 2022; Saraphol et al., 2025), suggesting a relatively stable core phylum-level structure. The relative abundances of major phyla did not differ significantly between HMUN and LMUN cows, suggesting that, under the shared diet and management conditions, the overall phylum-level composition of the fecal microbiota was broadly comparable between groups. At the genus level, however, distinct microbial signatures were evident. The HMUN cows were relatively enriched in *Christensenellaceae_R-7_group*, *Monoglobus*, *UCG-009*, *UCG-002*, and *Fibrobacter*, whereas LMUN cows showed relatively higher abundances of *Lachnospiraceae_NK3A20_group*, *Acetitomaculum*, *Blautia*, *Anaeroplasma*, and *Succinivibrio*, indicating that these taxa may serve as potential microbial markers associated with divergent MUN phenotypes. These fecal-based patterns are partly in line with previous rumen microbiome studies showing that bacterial community composition varies with MUN phenotype. Specifically, cows with low MUN or low MUN breeding values (MUNBV) were reported to harbor higher abundances of *Succinivibrionaceae_UCG_002* and unclassified members of *Ruminococcaceae*, whereas cows with high MUN or

high MUNBV were enriched in *Butyrivibrio*, *Lachnospiraceae_UCG-010*, *Desulfovibrio*, and *Lachnospiraceae_XPB1014* groups (Honerlagen et al., 2022, 2023). The consistent enrichment of *Succinivibrionaceae*-related taxa in low-MUN animals across both rumen and fecal samples suggests that microbial groups involved in efficient carbohydrate fermentation and nitrogen utilization are robustly associated with reduced urea excretion. In contrast, the predominance of certain *Lachnospiraceae* lineages and *Butyrivibrio* in high-MUN cows may reflect microbial configurations favoring greater ruminal ammonia production and, consequently, higher urea transfer to milk.

Spearman's correlation analysis further demonstrated that *Succinivibrio* and *Lachnospiraceae_NK3A20_group* were negatively associated with MUN. *Succinivibrio* produces succinate, which can be converted into propionate, a key gluconeogenic precursor and energy source in ruminants (Santos and Thompson, 2014; Wallace et al., 2015). Higher abundances of this genus have been linked to improved N conversion efficiency and lower methane emissions in cattle (Hernandez-Sanabria et al., 2012; Hailemariam et al., 2020; Xue et al., 2020; Li et al., 2022). Given that MUN decreases with increasing N utilization efficiency (Zhao et al., 2025b,c), the enrichment of *Succinivibrio* in LMUN cows observed in the present fecal analysis is consistent with its reported role in rumen-based studies. Similarly, *Lachnospiraceae_NK3A20_group*, which according to previous rumen studies is involved in fiber degradation as well as carbohydrate and protein metabolism and is commonly enriched in high-producing or high-feed-efficiency ruminants (Boeckaert et al., 2009; Paz et al., 2016; Rabee et al., 2025), was also negatively associated with MUN in feces, supporting its link with efficient energy and N utilization. However, most functional insights into these key taxa have been derived from rumen studies, and their specific metabolic roles in the lower gastrointestinal tract and their effect on host metabolism remain poorly characterized. Moreover, whether these microorganisms exhibit synergistic or antagonistic interactions in the intestine, or the mechanisms by which they influence MUN levels, remains unclear. These uncertainties highlight the need for further research in this field.

Fecal and plasma metabolomic profiling provides a useful approach for exploring the interplay between gut microbiota and host physiology, as these profiles reflect the metabolic activities of both microbial communities and the host. The KEGG pathway enrichment analysis indicated that metabolites differing between HMUN and LMUN cows were mainly involved in phenylalanine metabolism, purine metabolism, nucleotide metabolism, and α -linolenic acid metabolism. The HMUN cows showed relatively higher abundances of α -linolenic

acid-related metabolites, including traumatic acid and (9Z,11E,13S,15Z)-13-hydroxyoctadeca-9,11,15-trienoic acid, which were positively associated with *Christensenellaceae_R-7_group* and *UCG-009*, 2 taxa that were also correlated with higher MUN. Traumatic acid, an oxidized product of UFA, has been reported to possess antioxidant activity (Jabłońska-Trypuć et al., 2016); however, its physiological functions in the gastrointestinal tract of ruminants remain poorly characterized. The relative enrichment of α -linolenic acid-derived oxidative metabolites in HMUN cows may therefore suggest underlying differences in redox status and lipid-related microbial metabolism, which could be linked to a metabolic context characterized by less efficient N utilization and higher urea production, rather than indicating a direct causal contribution to MUN variation. In contrast, metabolites involved in phenylalanine metabolism, including dihydro-3-coumaric acid and 3-(2-hydroxyphenyl) propionic acid, were relatively more abundant in LMUN cows and were positively correlated with *Lachnospiraceae_NK3A20_group*, *Succinivibrio*, *Acetitomaculum*, and *Anaeroplasm*, taxa that were negatively associated with MUN. Phenylalanine is an EAA required for protein synthesis and the production of numerous bioactive compounds (Wu et al., 2014). The enrichment of phenylalanine-related metabolites in LMUN cows may suggest more efficient microbial transformation and host utilization of this AA, which could promote its incorporation into microbial and body protein rather than catabolism to ammonia, thereby contributing to reduced hepatic urea production and lower MUN levels.

Plasma metabolomic analysis further showed that metabolites discriminating HMUN and LMUN cows were mainly involved in lipid, AA, and carbohydrate metabolism. In LMUN cows, relatively higher concentrations of metabolites associated with β -alanine metabolism, D-AA metabolism, and glyoxylate and dicarboxylate pathways were observed, including D-pantothenic acid, DL-histidine, L-serine, and 4-hydroxy-2-oxoglutaric acid. Pantothenic acid, a precursor of CoA, plays a central role in fatty acid oxidation and pyruvate metabolism (Abo Alrob and Lopaschuk, 2014), and its elevated concentration may indicate enhanced central energy metabolism. Such metabolic features could support microbial protein synthesis and host anabolic processes, thereby improving the incorporation of ruminal ammonia into microbial and body protein and reducing the proportion of nitrogen entering the hepatic urea cycle. The metabolite, 4-hydroxy-2-oxoglutaric acid, serves as a precursor of hydroxyproline (Katsumata et al., 1997), an AA essential for structural protein synthesis and nitrogen homeostasis. It is also involved in the kidney-mediated production of glycine and other AA (Kaul et al., 2008), thereby interfacing with multiple metabolic pathways. The L-serine is

not only a proteinogenic AA but also plays roles in immune regulation and one-carbon metabolism (Tabatabaie et al., 2010; El-Hattab, 2016; Wu et al., 2020). Its higher abundance in LMUN cows may therefore reflect a more active AA metabolic network supporting protein synthesis, tissue turnover, and mammary secretion, which is metabolically compatible with improved nitrogen retention and lower circulating and milk urea concentrations. Collectively, the coordinated upregulation of energy- and AA-related metabolic pathways in LMUN cows may reflect a metabolic state more favorable for efficient nitrogen utilization, which could in turn contribute to lower hepatic urea synthesis and a reduced transfer of urea from blood to milk.

It should be emphasized that this study is exploratory, based on a single herd with a limited sample size ($n = 7$ per group), and provides correlational rather than causal evidence. Fecal metabolites and microbial profiles offer indirect insight into postruminal processes, and functional activities of specific microbial pathways were not directly measured. Therefore, the associations observed in this study between microbial taxa and MUN, as well as between metabolites and MUN, should be interpreted as system-level consistencies rather than as direct causal or mechanistic evidence. Future studies with larger populations, multiple herds, and integrative functional analyses are needed to validate these associations and clarify the mechanisms underlying MUN variation.

CONCLUSIONS

We identified differences in fecal microbial composition and modest variations in metabolite profiles between cows with high and low MUN levels. The HMUN cows were relatively enriched in *UCG-009*, *UCG-002*, and *Christensenellaceae_R-7_group*, whereas LMUN cows showed higher abundances of *Lachnospiraceae_NK3A20_group*, *Succinivibrio*, *Acetitomaculum*, and *norank_f_Bifidobacteriaceae*. A subset of fecal and serum metabolites showed numerical differences between groups and were mapped to related metabolic pathways. The *UCG-009* and *Christensenellaceae_R-7_group* were positively associated with MUN concentration; *Succinivibrio*, *Lachnospiraceae_NK3A20_group*, *Prevotellaceae_UCG-001*, *Anaeroplasm*, and *Acetitomaculum* were negatively correlated. Results indicate that variation in gut microbial communities and their metabolic activities is linked to divergent MUN phenotypes in dairy cows. Given the exploratory nature of the study, limited sample size, and cross-sectional design, these findings should be interpreted with caution. Further studies with larger cohorts and integrative metagenomic and metabolomic approaches are warranted to validate these associations and to elucidate the mechanisms by

which gut microbiota and their metabolic activities influence MUN phenotypes.

NOTES

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Nonstandard abbreviations used: ACE = angiotensin-converting enzyme; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ASV = amplicon se-

quence variants; FC = fold change; HMUN = high MUN; KEGG = Kyoto Encyclopedia of Genes and Genomes; LC = liquid chromatography; LDA = linear discriminant analysis; LefSe = linear discriminant analysis effect size; LMUN = low MUN; MUNBV = MUN breeding values; NCBI SRA = National Center for Biotechnology Information Sequence Read Archive; NEFA = nonesterified fatty acids; PC = principal component; PCA = principal component analysis; PCoA = principal coordinate analysis; PLS-DA = partial least squares discriminant analysis; VIP = variable importance in projection.

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